

Semantic Search: A Short Introduction

What semantic search is

Semantic search finds information by meaning rather than by exact words. A keyword search for 'car repair' misses a page that only says 'fixing an automobile', while a semantic search returns it because the two phrases mean nearly the same thing.

It works by turning both the documents and the question into vectors, lists of numbers that capture meaning, and then returning the documents whose vectors are closest to the question's vector.

How vector embeddings work

An embedding model is a neural network trained on large amounts of text so that sentences with similar meaning produce similar vectors. Each text becomes a point in a space with hundreds of dimensions; a small model such as bge-small produces 384 numbers per text.

Closeness is usually measured with cosine similarity, which compares the direction of two vectors. A score near 1 means the texts are about the same thing; a score near 0 means they are unrelated.

How AI learns from data

Machine learning systems learn patterns from examples instead of following hand-written rules. During training, the model makes a prediction, measures how wrong it was, and adjusts its internal parameters slightly to reduce that error. Repeating this over millions of examples lets the model generalise to inputs it has never seen.

Embedding models are trained this way on pairs of related texts, such as a question and its answer, so that related pairs end up close together and unrelated pairs end up far apart.

Chunking long documents

Embedding models read a limited amount of text at once, so long documents are split into chunks of a few hundred words. Neighbouring chunks usually overlap a little, so that a sentence cut at a boundary still appears whole in at least one chunk.

Each chunk is stored with its vector and with metadata such as the document name and page number, so search results can point back to their source.